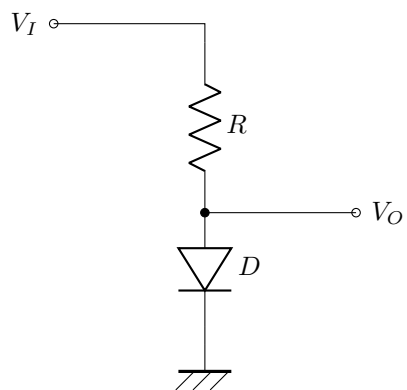
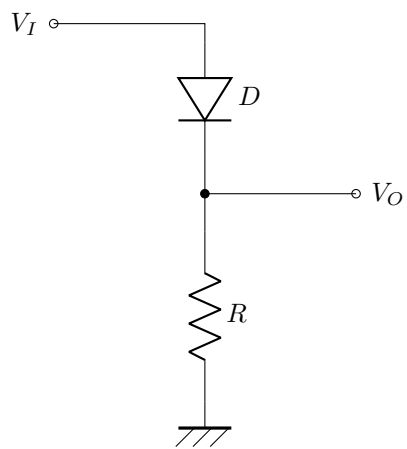


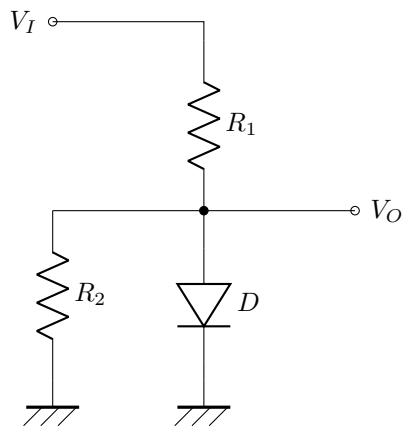
Example 1: diode and resistor I



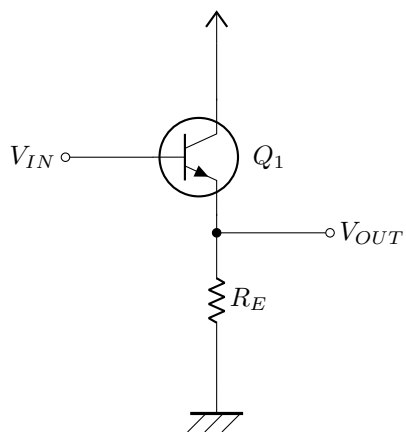
Example 2: diode and resistor II



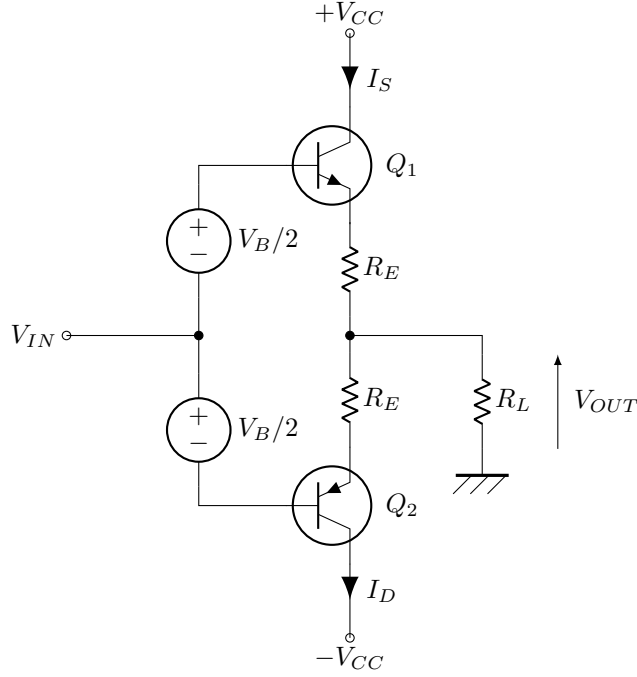
Example 3: 1 diode and 2 resistors



Example 4: common emitter



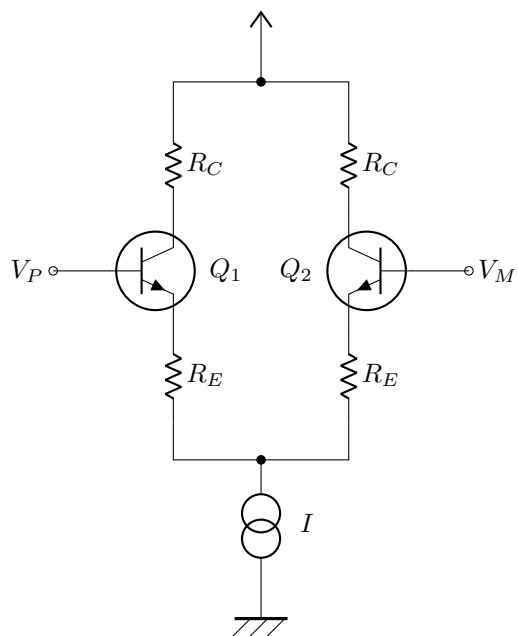
Example 5: class AB amplifier



Derivation Assuming for a BJT that $I_C = f(V_{BE})$ we get the following:

$$\begin{cases} I_S = \frac{a_S}{1+R_E a_S} (V_{IN} - V_{OUT} + \frac{V_B}{2}) + \frac{b_S}{1+R_E a_S} \\ I_D = \frac{a_D}{1+R_E a_D} (-(V_{IN} - V_{OUT}) + \frac{V_B}{2}) + \frac{b_D}{1+R_E a_D} \\ V_{OUT} = R_L (I_S - I_D) \end{cases}$$

Example 6: differential pair



Example 7: ring modulator

